

Lixin Xu

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RESEARCH INTERESTS

I am deeply committed to advancing the field of robotics with a focus on the integration of cognitive and athletic intelligence to create robotics foundation models that could revolutionize our modern industry and society.

EDUCATION

The Hong Kong University of Science and Technology <i>Guangzhou, China</i>	Starting Sept 2025
Ph.D Student in Robotics. Supervisor: Renjing Xu	
Georgia Institute of Technology <i>Atlanta, United States / Shenzhen, China</i>	Dec 2024
M.Sc. in Electrical and Computer Engineering	GPA: 4.0/4.0
Qingdao University <i>Qingdao, China</i>	June 2021
B.Sc. in Mechanical Engineering	GPA: 90.96 / 100 (Top 1%)

INTERNSHIP

The Hong Kong University of Science and Technology <i>Guangzhou</i>	June 2025 - Sept 2025
Research Assistant. Supervisor: Renjing Xu	
Tsinghua University	Jan 2025 - May 2025
Research Intern at Institute for AI Industry Research (AIR), Supervisor: Yilun Chen and Wenchao Ding	
National University of Singapore	June 2024 - Dec 2024
Research Intern at School of Computing, Supervisor: Prof. Lin Shao	

PUBLICATION

WHIRL: Intervention-Aware World Models with Real-World RL for Dexterous Manipulation <i>CoRL 2026</i>	Accepted
- Jiaju Yin, Zhenhui Zhang, Lixin Xu, Heng Zhang, Jun Shao, Yating Feng, Arash Ajoudani, Renjing Xu. - Website: https://whirl-dexterous.github.io/	
PDF-HR: Pose Distance Fields for Humanoid Robots <i>NeurIPS 2026</i>	Accepted
- Yi Gu, Yukang Gao, Yangchen Zhou, Xingyu Chen, Yixiao Feng, Mingle Zhao, Yunyang Mo, Zhaorui Wang, Lixin Xu, Renjing Xu. - Website: https://gaoyukang33.github.io/PDF-HR/	
DexFormer: Cross-Embodied Dexterous Manipulation via History-Conditioned Transformer <i>arXiv</i>	Preprint
- Ke Zhang, Lixin Xu, Chengyi Song, Junzhe Xu, Xiaoyi Lin, Zeyu Jiang, Renjing Xu. - Website: https://davidlxu.github.io/DexFormer-web/	
DexSinGrasp: Learning a Unified Policy for Dexterous Object Singulation and Grasping in Densely Cluttered Environments <i>IEEE Robotics and Automation Letters</i>	Accepted
Lixin Xu, Zixuan Liu, Zhewei Gui, Jingxiang Guo, Zeyu Jiang, Tongzhou Zhang, Zhixuan Xu, Chongkai Gao, Lin Shao. - Website: https://nus-lins-lab.github.io/dexsingweb/	

TALKS & WORKSHOPS

RevoSim: Scalable Multimodal Tactile Simulation for Dexterous Manipulation <i>IROS 2026</i>	Workshop Lighting Talks
- Ke Zhang, Lixin Xu, Ziyi Wang, Jie Tan, Chuanyu Li, Renjing Xu. - Website: sensact4dexman-workshop.github.io - Project page: davidlxu.github.io/RevoSim-web/	

PROJECT EXPERIENCE

- What Bimanuals Can Do (WBCD) Challenge** | *ICRA 2026* June 2026
- Champion. Team member on a bimanual manipulation system.
- What Bimanuals Can Do (WBCD) Challenge** | *ICRA 2025* May 2025
- Champion. Refer to technical report <https://arxiv.org/abs/2508.14542>
- Digital Twin-based Advanced Control of a Pneumatic Parallel Manipulator** | Jan. 2023 - Jan. 2024
- Kinematics and dynamics analysis, disturbance rejection control, nonlinear controllers and observers.
- An ORB-based Stereo Vision Odometry** | *Undergrad thesis* Jan 2021 - Jun 2021
- Designed PyVO (Python Visual Odometer), an ORB-based visual stereo odometer.
- The 6th National Engineering Training Competition - Material Handling Robot** | *Special Prize* Dec 2018 - Jun 2019
- Built a STM32-based Mecanum Wheel robot with OpenMV for object recognition.

HONORS & AWARDS

- National Scholarship** Ministry of Education of China 2021
- National Scholarship** Ministry of Education of China 2019

SKILLSET

Programming and tools Isaac Lab, Isaac Gym, Python/PyTorch, C/C++, MATLAB/Simulink, ROS/ROS2, Cmake, $\LaTeX$, Solid-works

ML & Decision Making Reinforcement Learning, Imitation Learning, VLA

Control & Actuation Nonlinear control, disturbance rejection control, dynamics modeling

Languages Mandarin(native), English(IELTS 7.5, GRE 322, CET6 614), Cantonese (colloquial), German(A2), Japanese(N4)